ChatGPT

Victaulic for Data Center Thermal and Fire Piping Systems

Executive summary

Victaulic has one of the most integrated publicly visible portfolios for data centers among mechanical-piping vendors: grooved pipe joining, fittings, flow control, equipment modules, hydronic balancing, and fire protection all sit under one supplier, and the company now markets an explicit mission-critical/data-center offer with multiple published data-center project references. In practical terms, that makes Victaulic unusually strong where a project wants one piping platform spanning chilled water or technology-cooling loops, branch balancing, seismic accommodation, phased expansion, and life-safety systems. [1]

The strongest public evidence for Victaulic in data centers is not a single "TCS catalog," but a combination of: an official data-center page naming "Technical Cooling System Equipment Modules," "Liquid-To-Rack Header Modules," and "Branch Lines & Valve Trains"; standardized commercial products such as the Series 250 butterfly valve family, Style 07 rigid coupling, Style 647 dielectric transition fittings, TA balancing valves, and FireLock / Vortex fire-protection products; and case studies showing installation in live or space-constrained data-center environments. [2]

The main limitation in public evidence is that Victaulic does **not** appear to publish product-level MTBF values or a clear public SKU list for the TCS module layer itself. The public documents reviewed here are much stronger on pressure ratings, materials, gasket chemistry, installation rules, certifications, and testing than on probabilistic reliability metrics or off-the-shelf module part numbers. Warranty language is usually incorporated by reference to the current price list rather than stated as a product-specific time period in the datasheets reviewed. [3]

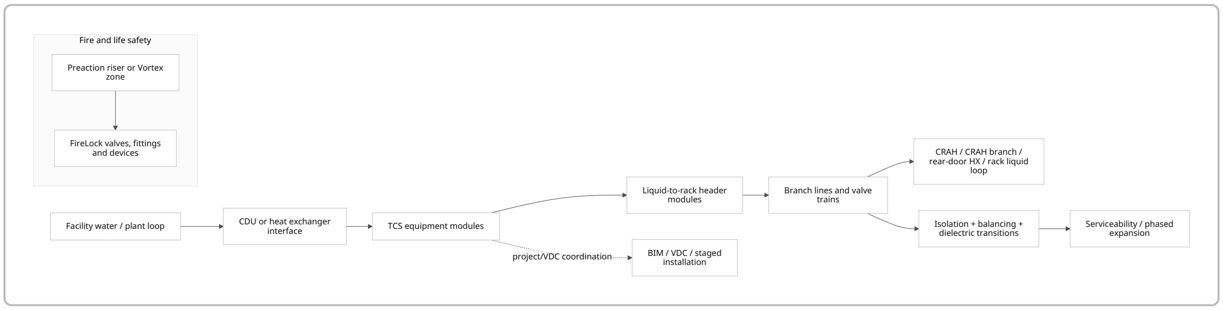
For selection, the most defensible default for a modern data center is: rigid grooved primary headers, flexible grooved joints only where movement is intentional, Series 250 valves matched to the actual pipe material, EPDM-family gaskets for water/glycol service within temperature limits, dielectric transitions wherever stainless/carbon and copper interfaces exist, and a separate fire-protection decision between nitrogen-assisted preaction and water/nitrogen hybrid suppression depending room-integrity and asset-protection goals. [4]

Scope and terminology

The user's question correctly flagged an ambiguity: Victaulic's public data-center page names "Technical Cooling System Equipment Modules," but the official public datasheets reviewed here do **not** provide a formal acronym expansion for "TCS." A secondary Victaulic LinkedIn snippet refers to a "Technical Cooling System (TCS) loop." On that basis, this report treats **TCS** as Victaulic's technology/technical cooling loop for liquid-cooling distribution from facility water or CDU interfaces toward rack rows and branch circuits.

Because the public module layer is not exposed with a standard catalog structure, this report focuses both on the named TCS module families and on the standard Victaulic parts that obviously support them. [5]

Victaulic's broader portfolio is relevant because the company's data-center value proposition is system-level, not only component-level. The official products taxonomy includes pipe joining, fittings, flow control, equipment modules, hydronic balancing, and fire-protection systems; the fire side includes sprinklers, couplings, fittings, valves/devices, and the Victaulic Vortex special-hazards line. [6]



The diagram above is a synthesis of Victaulic's public "Technical Cooling System Equipment Modules," "Liquid-To-Rack Header Modules," and "Branch Lines & Valve Trains" language, combined with the company's standard product families for flow control, balancing, and fire protection. [7]

Victaulic portfolio and TCS-relevant products

Victaulic's public data-center page is strategically important because it shows that the company is not pitching data centers only with generic grooved couplings; it is explicitly positioning purpose-built thermal-cooling assemblies. At the public-web level, though, the most specification-grade information remains concentrated in standard valve, coupling, fitting, balancing, support, and fire-protection documentation rather than a downloadable TCS module catalog. [8]

Family	Publicly visible series/style numbers	Key public specs	Why it matters in data centers	Primary source
TCS module layer	Technical Cooling System Equipment Modules; Liquid-To-Rack Header Modules; Branch Lines & Valve Trains	Official public page names these modules, but the sources reviewed did not expose standard sizes, pressure classes, or public part-number tables for them.	Indicates Victaulic is selling a module-based liquid-cooling distribution concept, likely engineered/ quote-driven rather than purely catalog-driven.	[5]

Family	Publicly visible series/style numbers	Key public specs	Why it matters in data centers	Primary source
Rigid grooved coupling	**Style 07 Zero-Flex**	1–12 in; vacuum to 750 psi; ductile iron housing to ASTM A536; orange enamel or hot-dip galvanized; Grade E EPDM or Grade T nitrile gasket options; support/hanging requirements tied to ASME B31.1, ASME B31.9, and NFPA 13.	Good fit for rigid mains, pump-room headers, and mechanical room spools where axial/angular movement should be restricted.	9
Flexible grooved coupling	**Style 77**	¾–24 in; up to 1000 psi; lead-free requirements per NCC Vol. 3 noted on product page.	Useful where designers intentionally want movement allowance for vibration, settlement, offsets, or thermal/seismic accommodation.	10
Stainless butterfly valve	**Series 250-S4**	2–12 in; stainless-steel pipe / OGS; up to 300 psi; body in ductile iron, stainless end faces/disc; fluoroelastomer blend seat, -10°F to 180°F; UL classified to NSF/ANSI/CAN 61 and 372; IAPMO/UPC; MSS SP-67; ASME B16.34 pressure-testing requirements.	Best public-fit Victaulic valve for stainless technology-cooling / chilled-water branches and header isolation.	11
Copper butterfly valve	**Series 250-C / 250-CU**	2–8 in; CTS copper; up to 300 psi; ductile-iron body; aluminum-bronze end faces/disc; fluoroelastomer blend seat, -10°F to 180°F; UL NSF/ANSI/CAN 61/372; IAPMO/UPC; MSS SP-67; ASME B16.34.	Strong fit where CDUs, manifolds, or branch hardware stay on copper tubing for compactness or contractor preference.	12

Family	Publicly visible series/style numbers	Key public specs	Why it matters in data centers	Primary source
Material transition fitting	**Style 647-TT / 647-GT / 647-GG**	647-TT ½–4 in; 647-GT 1–4 in; 647-GG 2–6 in; 300 psi; copper-silicon body to UNS C87850; certified to NSF/ANSI/CAN 61 and 372; intended to join carbon or stainless steel to copper tubing while maintaining minimal electric potential.	One of the most directly relevant data-center parts when stainless or carbon steel headers transition to copper branch or rack-adjacent tubing.	13
Pipe clamps / movement control	**A81 / A82 / A83 / A84 Pipe Clamp Anchors**	2–12 in; for Schedule 40 or greater carbon steel pipe; plate ASTM A36, structure tubing ASTM A500 Grade C, clamps ASTM A653 Grade 33; engineered to connect to concrete or steel support structure to direct piping movement and transfer loads.	Relevant for intentional movement control, especially in constrained data halls, mechanical mezzanines, and seismic zones.	14
Hydronic balancing / branch control	**TA Series TC / TCM**	½–1 in; up to 230 psi; -4°F to 250°F; TC for on-off control, TCM for modulating or on-off control.	Useful on terminal branches serving CRAHs, in-row coils, or hydronic heat exchangers where accurate branch control and commissioning discipline matter.	15
Fire-protection piping and fittings	**FireLock fittings**	1¼–12 in; ductile iron ASTM A536; designed exclusively for Victaulic fire-listed/approved couplings; UL, FM, LPCB, VdS, CE/CPR, UKCA visible on submittal.	Supports sprinkler / preaction system piping within a coordinated Victaulic fire ecosystem.	16

Family	Publicly visible series/style numbers	Key public specs	Why it matters in data centers	Primary source
Preaction / dry-system fire control	**FireLock NXT Series 769N**	1½–8 in; up to 300 psi; available bare, pretrimmed, as riser or cabinet system; Microsoft Taiwan case notes 13 psi minimum air pressure with Potter nitrogen generator; older Victaulic datasheet snippet states 600 psi factory hydrostatic test and 13 psi required air pressure.	Strong candidate for data-center preaction systems where low air pressure and reduced internal corrosion are valued.	[17]
Special-hazard suppression	**Vortex 1500 Hybrid Fire Suppression System**	Used in CDC project across multiple zones; CDC case states average 10-micron droplets, effective in open/naturally ventilated areas, no annual fan testing at that site, and 97% less water than high-pressure water mist systems.	Particularly relevant where owners prioritize minimized collateral damage to electronics and reduced room-integrity burden.	[18]

Two bottom-line observations follow from the table. First, the **module layer** is clearly present in Victaulic's public data-center narrative, but public **catalog transparency** is much stronger for the standardized valves, couplings, fittings, balancing valves, supports, and fire devices that sit around those modules. Second, for liquid-cooled data centers, the most consequential "specific parts" are often not exotic module SKUs but the interoperability parts: Series 250 valves, Style 647 transitions, gasket grades, anchors, and balancing/control valves. [19]

Installation and compatibility for data centers

Victaulic's public literature is strongest where data-center owners care about compressed schedules, tight spaces, and live-facility construction. The company explicitly markets grooved systems for confined spaces, and its data-center case studies repeatedly describe successful use where underfloor zones, mezzanines, elevator transport constraints, campus congestion, or active facilities made welding or large field assemblies unattractive. In the 1623 Farnam project, the contractor used shorter pipe sections that could be moved on elevators because the joining method reduced field-assembly burden; in Green Mountain DC-1, the underfloor cooling space drove installation and maintenance priorities; in IT4Innovations, offsite prefabrication and VDC were used to overcome a tiny installation footprint. [20]

For dynamic conditions, Victaulic's guidance is clear that a data-center system should not be "all flexible" by default. On the seismic side, Victaulic says most grooved couplings in a building system are typically rigid, with a smaller share flexible where movement accommodation is intended; on the seismic design-data publication, Victaulic states that rigid couplings such as Style 07 can be used where rigidity is needed, while flexible couplings help reduce stresses, damp vibration through the gasket, and accommodate directional offsets and movement between independently moving sections. That is a good fit for data centers, where pumps, chillers, CRAHs, and building-structure crossings all produce different movement problems. [21]

Material compatibility is another core data-center question because cooling loops commonly mix stainless steel, carbon steel, copper tubing, elastomers, water treatment chemicals, and sometimes glycol. Victaulic's Seal Selection Guide gives Grade E EPDM a general range of -30°F to 230°F and says it may be specified for hot-water service plus dilute acids, oil-free air, and many chemical services; Victaulic's own Q&A page also states that EPDM gaskets are compatible with glycol. Victaulic simultaneously warns that general ranges are not enough for final selection and directs engineers to the Gasket Chemical Services Guide for system-specific verification. That means a water/glycol data-center loop is usually feasible within Victaulic's EPDM-family envelope, but the **final check must be fluid-specific**. [22]

Butterfly-valve installation guidance is unusually relevant for data-center operators because serviceability and accidental-valve issues are high-consequence. Victaulic's Series 250 instructions say the connected piping must be properly supported so the joints are not subjected to bending or shear loads, that valves should be located and oriented to minimize dynamic torque on valve life, that the valve should not be used as a piping support, and that welding to the valve voids warranty. The same instructions say the valve accommodates up to 2 in of insulation, and recommend the disc be positioned no less than 30 degrees open for throttling service, with best results generally between 30 and 70 degrees open depending on system characteristics. [23]

A practical data-center interpretation is that Victaulic is especially attractive when the design team values **accessible joints, visible verification, modular spool replacement, and planned movement control**. The public sources reviewed do not show a dedicated Victaulic leak-detection product for the cooling loop, so the sensible design inference is to pair Victaulic's accessible, maintainable joints with **independent** facility leak-detection coverage at CDUs, manifolds, valve trains, and low points during the engineered design stage. That final leak-detection recommendation is an engineering inference rather than a Victaulic catalog feature. [24]

Design issue	Practical Victaulic approach	Why the evidence supports it	Source
Tight plant rooms / mezzanines / underfloor zones	Favor grooved assemblies and prefabricated spools over welded field work.	Victaulic explicitly markets confined-space installation; multiple data-center case studies cite spatial constraints and phased installation benefits.	25
Pump / equipment vibration	Use rigid mains with flexible joints only at intentional movement points.	Victaulic's seismic design data says flexible couplings reduce stress transmission and gasket damping attenuates vibration.	26
Seismic crossings / differential movement	Use seismic/movement design rather than indiscriminate flexible joints everywhere.	Victaulic says most grooved joints in a building system are rigid, with flexible elements used strategically.	27

Design issue	Practical Victaulic approach	Why the evidence supports it	Source
Mixed stainless/ copper systems	Use material-matched Series 250 valves plus Style 647 transition fittings.	Victaulic publishes separate stainless and copper valve families and a dedicated dielectric transition fitting.	28
Water/glycol chemistry	Start with Grade E/EHP EPDM if within temperature range, then verify exact chemistry.	Victaulic says EPDM is compatible with glycol, but final fluid compatibility must use GSG-100.	29
Serviceability / branch isolation	Keep valves accessible, independently support them, and avoid using valves as structure.	Series 250 instructions emphasize support, dynamic-torque minimization, and maintenance-safe handling.	30

Performance, reliability, and compliance

Victaulic's public reliability story in data centers is built around installation certainty, serviceability, and testing/certification rather than MTBF. On the mission-critical page, Victaulic says grooved systems can be visually verified for proper installation and that reliability is engineered into the design for the facility lifespan. That is reinforced by project case studies where data-center owners emphasize fast phased expansion, no hot work, reduced downtime, low-pressure nitrogen-supported fire systems, or minimal water residue. [31]

On compliance, Victaulic's commercial-cooling valves are well documented. The Series 250-S4 and 250-C both carry UL classification to NSF/ANSI/CAN 61 and 372 up to commercial hot-water rating, are certified by IAPMO R&T to the Uniform Plumbing Code and MSS SP-67, and state compliance with ASME B16.34 Section 7 pressure testing. Style 07 shows a certification/listing block including cULus, FM, LPCB, VdS, and CE/CPR on the submittal screenshot, while FireLock fittings show UL, FM, LPCB, VdS, CE/CPR, UKCA, and CCC marks on the fire submittal. Victaulic's fire-protection page also states that the company has more than 600 fire certificates worldwide for over 2,000 fire-product models. [32]

Failure modes are described indirectly but clearly in the documents. Victaulic warns that gasket misuse with incompatible fluids or temperatures can reduce performance and service life; Grade E EPDM is not compatible with petroleum or steam service, and Grade T nitrile is not for hot-water or steam service. On the valve side, Victaulic warns against incompatible materials, improper support, bending/shear loads, direct welding, and end-cap pressurization hazards. These documents are useful because they map to real data-center failure modes: wrong coolant-to-elastomer pairing, unrestrained vibration or bending at valve bodies, and maintenance accidents during isolated-branch work. [33]

What is **missing** from the public documents reviewed is just as important. I did **not** find product-level MTBF or MTTF values for Victaulic couplings, valves, fittings, or TCS modules, and I did **not** find publicly stated warranty durations in the core datasheets reviewed. Instead, datasheets repeatedly say to refer to the current price list or contact Victaulic for warranty details. For a data-center procurement package, that means reliability diligence should lean on certification, installation discipline, material compatibility, factory and code testing, and project references rather than waiting for an OEM MTBF line item that seems not to be publicly published. [34]

Topic	What is publicly available	What is not clearly public in the sources reviewed	Source
Pressure / temperature / materials	Published for Style 07, Series 250-S4, Series 250-C, Style 647, TA TC/TCM, FireLock 769N.	Public TCS-module specification sheets were not found.	35
Testing	ASME B16.34 pressure-testing references on Series 250; Style 769 hydrostatically tested to 600 psi in Victaulic fire datasheet snippet; multiple UL/FM/NSF/IAPMO listings.	No public field failure-rate database.	36
Reliability metric	Installation visual verification and long-life marketing language.	No public MTBF/MTTF values found.	37
Warranty	Warranty referenced through current price list / Victaulic contact.	No standardized duration stated in the reviewed datasheets.	34

Data-center project evidence

Victaulic's published project evidence matters because it shows where the company has actually been used, not only what it claims it can do. The cases below collectively cover cooling loops, balancing, preaction fire systems, and hybrid fire suppression in North America, Europe, Asia, and Australia. 38

Project	What Victaulic was used for	Why it is analytically relevant	Source
1623 Farnam	Chilled-water and air-cooling systems; Victaulic grooved solutions; Series 761 Vic-300 MasterSeal butterfly valves; differential-pressure balancing valves.	Good evidence for live-facility retrofit logic: no hot works, shorter pipe sections moved by elevator, phased implementation, and branch balancing for floor-by-floor activation.	39
Green Mountain DC-1 Stavanger	Grooved couplings, fittings, and valves on cooling pipes carrying fjord water; underfloor cooling environment.	Strong evidence for maintainability and scalability in confined spaces; owner explicitly cited phased growth flexibility and easy maintenance.	40
IT4Innovations	VDC + prefabricated cooling pipework; rigid and flexible grooved couplings; DN20 to DN200 system routing to roof cooling towers.	Useful case for dense prefabrication, small-site logistics, and mixed rigid/flexible design on real cooling infrastructure.	41

Project	What Victaulic was used for	Why it is analytically relevant	Source
43,000 sq ft Northern California data center	Chiller plant and variable-flow split system; 14–24 in piping; compressed three-month delivery.	Shows Victaulic on large-diameter mechanical plant with built-in redundancy and future expansion requirements.	42
Microsoft Taiwan data center	Complete fire-protection package including sprinklers, butterfly valves, gate valves, deluge and preaction valves, couplings and fittings; FireLock NXT 769N with Potter nitrogen generator at 13 psi minimum air pressure.	Important evidence that Victaulic can deliver a full fire package for hyperscale-class data-center projects and that distributor inventory / air freight can be used tactically to protect schedule.	43
Canberra Data Centres	Victaulic Vortex 1500 hybrid fire suppression across multiple zones.	Best evidence for special-hazard positioning in ICT rooms: 10-micron droplets, open naturally ventilated compatibility, no room sealing, no annual fan testing at that site, and 97% less water than high-pressure water mist.	18

The most important pattern across the case studies is that Victaulic wins less on marginal component performance than on **system constructability and lifecycle flexibility**: no hot work, faster phasing, easier branch isolation, prefabrication, mixed-material transitions, and better fit in spaces where shutdown risk is expensive. That is a particularly good match for both retrofit colocation work and newer liquid-cooling builds where rack density may evolve after day one. [44]

Procurement, competitor landscape, and supply chain

Victaulic's current U.S. pricing page is live and exposes a 2026 General Price List, multiple addenda, a dedicated fire-protection price list, and a distributor locator for U.S. and non-U.S. markets. Public distributor coverage in the sources reviewed includes Ferguson, Porter Pipe, Macomb Group, Central Supply, and Sheret, which is a useful sign that common Victaulic parts are broadly channel-supported even when stock visibility is branch-specific or login-gated. [45]

Procurement and pricing signals

Item	Example part number or family	Public pricing signal	Practical procurement interpretation	Source
Style 07 rigid coupling	L012007PE0, L030007PE0, L030007GE0	Porter Pipe snippets show **$65.24** for 1¼ in L012007PE0 and **$115.75** for 3 in L030007PE0; a separate listing shows **$146.42** for galvanized 3 in L030007GE0.	Small / medium rigid couplings look like stocked-distribution items with double-digit to low-hundreds-dollar unit pricing depending size and finish.	46
Style 647 dielectric transition	F020647CTT	Porter Pipe snippet shows **$182.99** for a 2 in F020647CTT; Central Supply shows the same part as stocked but with login-gated price.	Transition fittings are usually not "commodity cheap" but are still standard catalog items, not custom assemblies.	47
Series 250-S stainless butterfly valve	V020250HP2, V060250HP6	Porter Pipe snippet shows **$1,831** for 2 in V020250HP2 and **$8,211** for 6 in V060250HP6; Sheret shows a 2 in 250-S4 listing at **$885.88** clearance.	Public pricing varies materially by channel, operator package, region, customer tier, and sale status; real bid pricing needs distributor quotation.	48
Series 250-CU copper butterfly valve	Series 250-CU addendum	Official Victaulic addendum lists **bare valve $1,000–$5,000** for 2–8 in; **lever-handle $1,228.50–$5,553**; **gear operator $2,012–$6,750**; **gear operator with chain wheel $2,492–$8,999**.	This is the clearest official public pricing evidence in the dataset and confirms that copper-system butterfly valves are distinctly higher-value, operator-dependent items.	49
TCS header / valve-train modules	Module layer	No public unit prices identified in the reviewed sources.	Treat as engineered / quote-driven scope and engage Victaulic plus distributor early.	50

The public sources reviewed do **not** support a single reliable day-based lead-time range. What they do show is that common couplings and fittings are widely distributed, while availability is often branch-specific or login-gated; specialized valves are more likely to require quote confirmation; and for schedule-critical data-center work Victaulic has explicitly built distributor inventory and even used air freight to keep projects on track. In other words, the public evidence supports a procurement model of **pre-buying long-lead valves/**

modules and relying on distributor stock for standard joining hardware, but not a universal lead-time promise in days or weeks. [51]

Competitor comparison

Vendor	Public portfolio breadth	Data-center evidence	Representative product evidence	Where it appears stronger	Where Victaulic still looks stronger	Source
Victaulic	Pipe joining, fittings, flow control, equipment modules, hydronic balancing, fire protection, Vortex special hazards.	Explicit mission-critical/data-center pages and multiple project case studies.	Series 250 valves, Style 07/77, Style 647, TA balancing, 769N, Vortex.	Integrated single-vendor system architecture and project support.	N/A baseline row.	[52]
Gruvlok / ASC Engineered Solutions	Pipe joining systems, flow control, fire/fabrication, support systems, performance products; ASC also has a data-center solutions page.	Yes—ASC has a data-center solutions page emphasizing availability and distribution reliability.	Gruvlok Series 8700 offset-disc butterfly valve for commercial grooved-end systems 2–8 in; ASC markets it as a high-pressure, low-torque design.	Broad construction-channel reach, strong distribution messaging, and credible grooved/valve competition.	Victaulic's public evidence is broader in mission-critical modules, hydronic balancing, and special-hazard fire systems.	[53]

Vendor	Public portfolio breadth	Data-center evidence	Representative product evidence	Where it appears stronger	Where Victaulic still looks stronger	Source
SPF / Anvil	Complete range of grooved products under ASC; brand positioning emphasizes quality + value; product set includes rigid couplings, fittings, valves.	Limited project-specific data-center evidence in the reviewed sources.	SPF/Anvil C-4 rigid coupling; SPF/Anvil range of grooved products; internationally sourced value line messaging.	Value-oriented grooved-product alternative, especially for fire/ commercial packages.	Victaulic appears stronger where owners want public evidence of mission-critical specialization and integrated fire/cooling solutions.	54
Tyco / Johnson Controls	Broad fire-protection valves, devices, corrosion solutions, sprinklers, and special valves.	Strong on fire protection, but the reviewed sources did not show a broad mechanical/ HVAC grooved-system + equipment-module story comparable to Victaulic's.	TYCO BFV-300/ BFV-300C grooved-end butterfly valve; Tyco valves/ devices/ components and corrosion-solutions pages.	Fire-only scope, especially where a project wants established sprinkler/ valve ecosystems.	Victaulic looks stronger when the owner wants one vendor spanning cooling-loop mechanics and fire systems together.	55

The practical conclusion is that **Gruvlok is the closest broad mechanical competitor**, **SPF/Anvil is a credible value competitor**, and **Tyco/JCI is a major fire-protection competitor rather than a true like-for-like mission-critical mechanical system competitor**. If a project's center of gravity is the cooling loop and system constructability, Victaulic's public evidence is stronger. If the center of gravity is fire protection only, Tyco/JCI remains highly relevant. [56]

On supply chain, Victaulic currently says it helps customers in more than 140 countries with **over 50 international facilities** and **more than 6,000 employees worldwide**. Current official/public documentation and certificates show headquarters in Easton, Pennsylvania, and production evidenced in Alburtis, Pennsylvania; Drezdenko, Poland; and Dalian, China; a Victaulic press release adds Chihuahua, Mexico as a facility producing ductile-iron castings and large-diameter products up to 72 in. I have not found one official page that appears to be a complete, current global manufacturing master list, so this should be read as **evidenced sites, not necessarily an exhaustive production map**. [57]

Recommendations and open questions

For a data-center selection memo, my recommendation is to treat Victaulic as a **system-platform candidate**, not just a coupling vendor. The best fit is a project that values staged deployment, prefabrication, constrained-space installation, mixed stainless/copper interfaces, and a coordinated fire strategy. If that is the owner brief, Victaulic's public evidence is materially stronger than a generic "grooved products" pitch. [58]

Recommendation	Why it is the most defensible choice from the evidence	Source
Use **rigid grooved mains** and reserve flexible joints for intentional movement points.	Victaulic's own seismic and movement guidance points to mostly rigid systems with flexible accommodation used strategically, not indiscriminately.	59
Match valve family to pipe material: **250-S4 for stainless**, **250-C / 250-CU for copper**, and use **Style 647** at dissimilar-metal interfaces.	Victaulic publishes separate stainless and copper valve families plus a dedicated dielectric transition fitting.	28
Default water/glycol loops to **Grade E or EHP EPDM** only after chemical confirmation.	EPDM has the right general service profile for water/glycol, but Victaulic explicitly requires GSG-100 verification for exact compatibility.	29
Independently support all butterfly valves and do not use them as piping supports.	Victaulic directly ties valve life and safe operation to proper support, minimized dynamic torque, and load control.	30
Pre-buy long-lead valves and module assemblies, but lean on distributors for common couplings/fittings.	Distributor evidence suggests broad catalog coverage for standard parts, while specialty valves/modules are less transparently stock-based and more quote/schedule-sensitive.	60
For fire protection, evaluate **769N preaction** versus **Vortex hybrid suppression** as separate decision paths.	The public evidence supports 769N where low-pressure nitrogen-supported preaction is attractive, and Vortex where electronic-asset protection and room-integrity burdens dominate.	61

Open questions and limitations remain. The largest one is still the TCS acronym and module-catalog visibility: public Victaulic sources reviewed here identify the module families, but they do not expose a formal public part-number/spec sheet structure for them. Before final procurement or basis-of-design decisions, I would specifically ask Victaulic or its distributor for: the exact TCS module bill of materials; the target coolant chemistry and temperature envelope; the intended pipe materials; the seismic design basis; the service strategy around leak detection and branch isolation; and the exact fire philosophy required in the white space, equipment rooms, and underfloor zones. Those answers will determine whether Victaulic's published standard parts are sufficient as-is or whether the project depends on engineered-to-order module content not publicly documented in the sources reviewed here.

---

1 6 52 https://www.victaulic.com/
https://www.victaulic.com/

2 5 7 8 19 50 Data Centers
https://www.victaulic.com/mission-critical/data-centers/?utm_source=chatgpt.com

3 31 37 https://www.victaulic.com/mission-critical/data-centers/
https://www.victaulic.com/mission-critical/data-centers/

4 9 34 35 https://assets.victaulic.com/assets/uploads/literature/06.02.pdf
https://assets.victaulic.com/assets/uploads/literature/06.02.pdf

10 https://www.victaulic.com/products/style-77-flexible-coupling/
https://www.victaulic.com/products/style-77-flexible-coupling/

11 28 32 36 https://assets.victaulic.com/assets/uploads/literature/17.69.pdf
https://assets.victaulic.com/assets/uploads/literature/17.69.pdf

12 https://assets.victaulic.com/assets/uploads/literature/22.69.pdf
https://assets.victaulic.com/assets/uploads/literature/22.69.pdf

13 https://assets.victaulic.com/assets/uploads/literature/22.21.pdf
https://assets.victaulic.com/assets/uploads/literature/22.21.pdf

14 https://www.victaulic.com/products/victaulic-pipe-clamp-anchors-nos-a81-a82-a83-and-a84/
https://www.victaulic.com/products/victaulic-pipe-clamp-anchors-nos-a81-a82-a83-and-a84/

15 https://www.victaulic.com/products/ta-series-tcm-tbv-ta-series-tc-tbv-terminal-balancing-and-control-valve-for-control/
https://www.victaulic.com/products/ta-series-tcm-tbv-ta-series-tc-tbv-terminal-balancing-and-control-valve-for-control/

16 https://assets.victaulic.com/assets/uploads/literature/10.03.pdf
https://assets.victaulic.com/assets/uploads/literature/10.03.pdf

17 https://www.victaulic.com/products/firelock-nxt-series-769n-preaction-system-check-valve/
https://www.victaulic.com/products/firelock-nxt-series-769n-preaction-system-check-valve/

18 https://www.victaulic.com/projects/canberra-data-centres-cdc-project/
https://www.victaulic.com/projects/canberra-data-centres-cdc-project/

20 25 https://www.victaulic.com/confined-space-installations/
https://www.victaulic.com/confined-space-installations/

21 27 59 https://www.victaulic.com/accommodating-seismic-movement/
https://www.victaulic.com/accommodating-seismic-movement/

22 29 33 https://assets.victaulic.com/assets/uploads/literature/05.01.pdf
https://assets.victaulic.com/assets/uploads/literature/05.01.pdf

23 30 https://assets.victaulic.com/assets/uploads/literature/I-250.pdf
https://assets.victaulic.com/assets/uploads/literature/I-250.pdf

24 https://www.victaulic.com/simplified-maintenance/
https://www.victaulic.com/simplified-maintenance/

26 https://assets.victaulic.com/assets/uploads/literature/26.12.pdf
https://assets.victaulic.com/assets/uploads/literature/26.12.pdf

38 39 44 58 https://www.victaulic.com/projects/1623-farnam/
https://www.victaulic.com/projects/1623-farnam/

40 https://www.victaulic.com/projects/green-mountain-dc-1-stavanger/
https://www.victaulic.com/projects/green-mountain-dc-1-stavanger/

41 https://www.victaulic.com/projects/it4innovations-data-center/
https://www.victaulic.com/projects/it4innovations-data-center/

42 https://www.victaulic.com/projects/43-thousand-square-foot-northern-california-data-center-project/
https://www.victaulic.com/projects/43-thousand-square-foot-northern-california-data-center-project/

43 61 https://www.victaulic.com/projects/microsoft-taiwan-data-center-project/
https://www.victaulic.com/projects/microsoft-taiwan-data-center-project/

45 https://www.victaulic.com/pricing/united-states/
https://www.victaulic.com/pricing/united-states/

46 https://www.porterpipe.com/1026392/product/n/victaulic-l012007pe0
https://www.porterpipe.com/1026392/product/n/victaulic-l012007pe0

47 https://www.porterpipe.com/1010914/product/n/victaulic-f020647ctt
https://www.porterpipe.com/1010914/product/n/victaulic-f020647ctt

48 https://www.porterpipe.com/1309574/product/n/victaulic-v020250hp2
https://www.porterpipe.com/1309574/product/n/victaulic-v020250hp2

49 https://assets.victaulic.com/assets/uploads/price_list/US/PL2026-GEN-ADD1.pdf
https://assets.victaulic.com/assets/uploads/price_list/US/PL2026-GEN-ADD1.pdf

51 60 https://www.porterpipe.com/1032388/product/n/victaulic-l030007pe0
https://www.porterpipe.com/1032388/product/n/victaulic-l030007pe0

53 https://www.asc-es.com/
https://www.asc-es.com/

54 https://www.asc-es.com/brand/spf-anvil
https://www.asc-es.com/brand/spf-anvil

55 https://docs.johnsoncontrols.com/tycofire/api/khub/documents/pFzo~EFuebkXFlwT5t7LZQ/content
https://docs.johnsoncontrols.com/tycofire/api/khub/documents/pFzo~EFuebkXFlwT5t7LZQ/content

[56] https://www.asc-es.com/solutions/data-centers
https://www.asc-es.com/solutions/data-centers

[57] https://www.victaulic.com/find-location/
https://www.victaulic.com/find-location/